

Module 7: Infectious Disease

Causes of Infectious Disease - How are diseases transmitted?

- describe a variety of infectious diseases caused by pathogens, including microorganisms, macroorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including:

Disease - any condition which interferes with normal functioning of the body

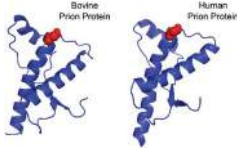
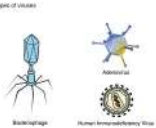
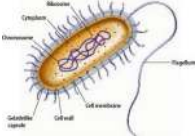
Health - complete state of physical, mental, social, emotional and spiritual wellbeing, not just absence of mortality

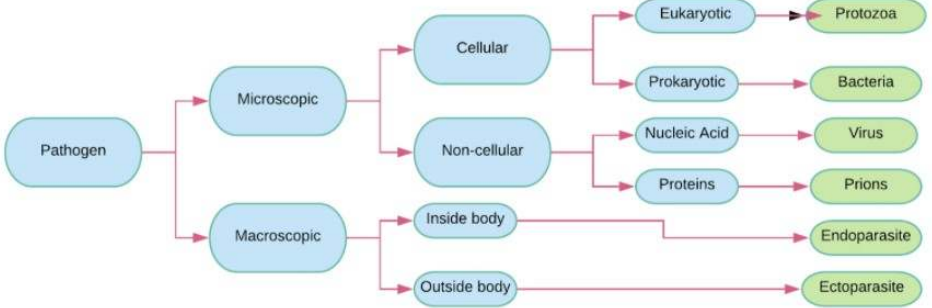
Pathogen - a disease causing agent or organism such as bacteria, viruses, fungus, protozoa, prions and macro parasites

Infectious diseases - caused by pathogens and can be transmitted from one person to another

Non-infectious diseases - not caused by pathogens but by other factors such as diet, environment, psychological issues and genetics. They cannot be transmitted from one person to another except through genetic inheritance.

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- classifying different pathogens that cause disease in plants and animals

Pathogen	Description	Example
Prion (non-living)	- Abnormally folded proteins made by DNA - Can make other proteins abnormal - Small and difficult to sterilise - Both infectious and hereditary 	Animal: Mad cow disease Plant: Chronic wasting disease
Virus (non-living)	- Package of DNA/RNA surrounded by proteins - Extremely small Non-cellular and therefore non-living + can't grow or reproduce - Invades host cells -> injects its genetic material into host cell -> cell metabolises new viruses - Can only survive in live cells - Vaccines are effective in getting rid of viruses 	Animal: HIV = AIDs, influenza = flu Ebola virus = ebla Plant: Tomato spotted wilt virus
Bacteria	- Tiny prokaryotic single cellular (no nucleus or membrane bound organelle) cell wall and cell membrane - Single strand of DNA - Reproduce through binary fission - Invade cells and form colonies that disrupt normal cell function - Can survive outside a host cell and in reservoirs through spores, therefore are hard to eliminate and therefore is hard to eradicated 	Animal: Tetanus, Tuberculosis, Bacillus, Lyme Disease Plant: Apple fire blight citrus canker
Protozoa	- Single celled eukaryotic organisms (cell membrane but no cell wall) - Free living heterotrophs (absorb nutrients from their host) - Form colonies that disrupt normal cell function - Classified by their method of locomotion: flagellates, pseudopods, ciliates, sporozoa 	Animal: Malaria, Flagellate Plant: downy mildew
Fungi	- Simple organism that are eukaryotic - multicellular or unicellular - Reproduce asexually using spores - Similar to plants but without chlorophyll	Animal: Foot tinea, Thrush, athletes foot Plant: black spot,

	- Can grow on skin and mucous membrane in body - Contain cell walls and membrane	mildew
Macro-parasites	- Multicellular parasites that obtain nutrients from their host and can cause disease and weight loss - Act as a vector for other pathogens - Has cell wall and membrane	Animal: Ticks, Tapeworm Plant: Dodder
 <pre> graph LR Pathogen --> Microscopic Pathogen --> Macroscopic Microscopic --> Cellular Microscopic --> Non-cellular Cellular --> Eukaryotic Cellular --> Prokaryotic Eukaryotic --> Protozoa Prokaryotic --> Bacteria Non-cellular --> Nucleic Acid Non-cellular --> Proteins Nucleic Acid --> Virus Proteins --> Prions Macroscopic --> Inside_body[Inside body] Macroscopic --> Outside_body[Outside body] Inside_body --> Endoparasite Outside_body --> Ectoparasite </pre>		

- investigating the transmission of a disease during an epidemic

Epidemic: sudden spike in the prevalence of an infectious disease in a region / community

Pandemic: sudden spike in a disease across the entire world

Enzootic: endemic amongst animals in a particular region

Example: Ebola Virus pathogen causing ebola fever

Severe and fatal disease that affects humans causing severe bleeding, organ failure and can lead to death. The rapid spread was due to lack of public health services and the socioeconomic status of the country = poor hygiene practices, infrastructure and lack of research.

Source:

2 year old boy in Guinea who was infected by wild bats

Virus spread from person to person to West African Countries: Guinea, Sierra Leone, Liberia

Incidence: 28,000 cases & 11,500 deaths

Mode of transmission: direct - The virus is transmitted to people from wild animals (such as fruit bats, porcupines and non-human primates) and then spreads in the human population through direct contact with the blood, secretions, organs or other bodily fluids of infected people and indirect contact with contaminated items.

Management of the outbreak:

Prevention: methods of reducing the chance of contracting the pathogen

Control: management of existing disease to stop its spread

- Ban on funeral and burial practices (cleaning of the dead) to avoid physical contact
- Specially designed ebola treatment hospitals
- Isolating cases and tracing contacts then isolating and monitoring them for 21 days
- Treated through replacement of body fluids to minimise symptoms
- Screening of passengers as domestic and international airports
- Isolation zones

- design and conduct a practical investigation relating to the microbial testing of water or food samples

Aim: To compare fresh and old food in terms of number and type of microbial colonies.

Hypothesis: Plates inoculated with old food will have a greater number and greater variety of different colonies growing on them than plates inoculated with fresh food.

Independent variable: age of food

Dependent variable: the number of microbial colonies growing on the plate and the number of different types of colonies found on the plate

Control: agar plate with no food

(This was to verify that microbes on plates come from food samples and are not present in the agar to begin with. This therefore validates the experiment by ensuring microbes grown are directly from the food)

Variables kept constant:

- Type of agar
- Sterile techniques
- Incubation time and temperature

Equipment

- Variables (source of reservoir)
- Inoculating loop
- Agar plates
- Bunsen burner

Risk assessment

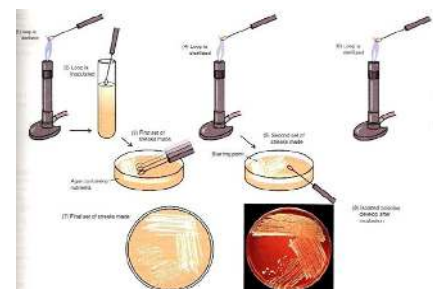
Identification of specific risk	Description of potential harm	Strategies to minimise risk
Agar plates with microbe colonies	Exposure to colonies could cause serious disease	-Always seal off Petri dishes after first exposure; -never open again; -sterilise after use and dispose of correctly
Sterilising inoculating loop	Burns caused by hot wire after heating inoculating loop	Ensure that the only part of the inoculating loop that is touched is the handle
Exposure to microorganisms	These microorganisms could cause disease	-Benches wiped with disinfectant; -use disposable gloves and wash hands with soap

Method:

1. Swab down your bench with a disinfectant solution.
2. Collect your inoculating loop (wand) and 10 sterile agar plates.
3. Practice various aseptic techniques that your teacher shows you. This includes correct sterilisation of your inoculating loop, correct streaking procedure, the need for re-sterilisation of the loop in between streaks, opening agar plates for minimum time necessary and at a small angle.
4. Streak your plates according to the following specifications (foods old vs fresh) + control group

Microbes in food set up:

5. Seal every plate with sticky tape, and label the food used. Please leave a one centimetre gap on the edge of the plate where there is no sticky tape, as the plate needs a small space for condensation to escape.
6. Incubate plates at 37 degrees Celsius for three days. After this time remove from the incubator and record your results.
7. Disinfect the working surface
8. Repeat experiment multiple times recording results in a table



Results: (full results in booklet)

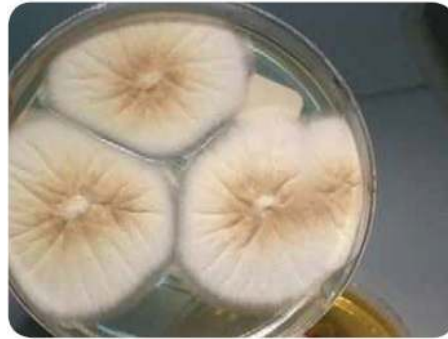
Bacterial colonies: normally shiny with a round uniform shape

Results: White, translucent, yellow, shiny, yellow translucent



Fungal colonies: normally fuzzy/fluffy or powdery in a splat like shape

Results: fluffy white, fluffy green



Conclusion: petri dishes inoculated with fresh doof samples grew fewer microbial colonies and fewer types of microbes than old/past use by date food samples. This means that fresh food is safer to eat than old food.

- investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission

Transmission - passing of pathogen from an infected person or reservoir to a non-infected person

Host - organism susceptible to a disease

Vector - carries a pathogen and transmits this pathogen

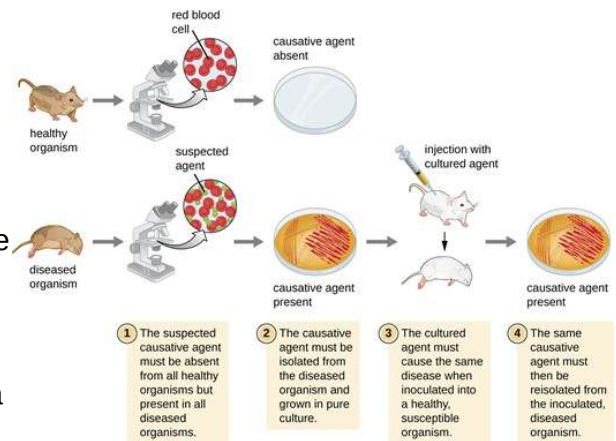
Direct Contact	Indirect Contact -----> Vector Transmission (type of indirect contact)	
Transfer of a pathogen via exposure to infected skin or secretions. Occurs when there is physical contact between the host and a non-infected organism. Vertical transmission: from mother to fetus (placenta, vaginal birth, breast feeding) Horizontal transmission: from one individual to another	Transfer of the pathogen to a new host via non-living objects or a living vector. Does not involve contact between host and another organism	Disease vectors are not pathogens themselves, rather they carry the pathogens that cause the disease. The pathogen is usually injected directly into the host by the vector, which overcomes the skin barrier. Malaria is caused by a protist parasite Plasmodium, which is carried by mosquitoes in their saliva. In this example, the Plasmodium is the pathogen and the mosquitoes are the vector. E.g. malaria caused by plasmodium protozoa on the vector mosquitoes.
Physical contact includes: · touching · sexual contact · kissing · contact with nasal or oral secretions · biting · direct contact with any blood or other body fluids (includes needles) · direct contact with wounds · prenatal (before birth or during pregnancy) or perinatal (around the time of birth) transmission.	Indirect means of transmission include: · airborne transmission — coughing or sneezing (droplets can travel up to 8 metres through the air) · touching an infected surface · contaminated food or water · infected surgical instruments · vectors such as mosquitoes, ticks and fleas.	

- investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:
 - Koch's postulates

Koch's Postulates (reasoning) was a method of proving which microorganisms caused which disease. He set up a criteria that must be met before a disease can be positively linked to a pathogen.

1. In all diseased organisms, the suspected pathogen must be found in abundance
2. Isolate this pathogen and grow in pure culture (not contaminated with other pathogens)
3. Infecting a healthy test subject with the suspected pathogen must result in the development of the same signs and symptoms as the identified disease
4. The pathogen must be re-isolated from the new host and be identical to the pathogen found in step 2

(Note: Koch was partially wrong as many people can carry a pathogen but not express symptoms of the disease)



Explanation for the transmission of infectious diseases:

Organisms must be infected by the same pathogen for the same disease to arise.

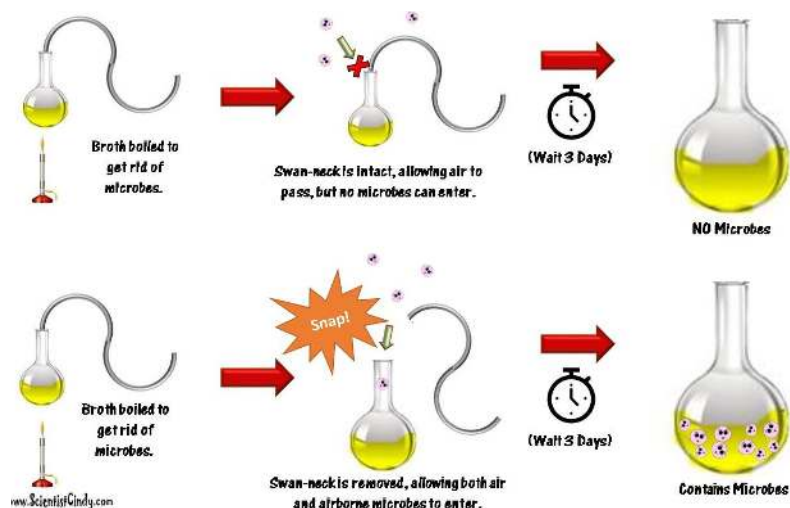
Found a causal relation between pathogen and development of an infectious disease.

– Louis Pasteur's experiments on microbial contamination

In the 19th century Pasteur found that fermentation was due to the presence of living microorganisms (bacteria) called yeast. Therefore he hypothesised that spoilage could be avoided by killing bacteria present through heat. **He proved this theory through his swan neck experiment.**

Argued germs (micro-organisms in the air) are the cause of disease, disputing the theory of spontaneous generation = **germ theory of disease**. (microbes come from pre-existing microbes) This understanding has led to treatment and storage of food to kill existing microbes and sealing of the food to prevent entry of new microbes. This has also reduced the incidence of food poisoning.

Pasteurisation = the process of heating food to kill pathogens



Risk Assessment: (RIP - Risk Injury Prevention)

Risk	Precaution/Prevention	Management	Disposal
Hot plate causing burn	Don't touch hot plate wait for it to cool down before putting away	Seek first aid	

Mould/bacteria growing on broth	Don't open the flask. Disinfect if spills occur	If exposed seek medical help	Lab assistant will destroy mould with heat/disinfectant
Glassware causing cuts	Place in the centre of the table and wear enclosed leather shoes & goggles to protect eyes	Seek first aid	Use broom + dustpan to sweep up glass and place into glass bin

Aim: To demonstrate that no decomposition will occur if the microbes are prevented from reaching the broth

Independent variable: shape of the flask neck (bent or straight)

Dependent variable: clarity of the broth

Controlled variables:

- Volume of broth
- Type of broth
- Boiling time
- Time the broth clarity is checked

Method:

1. Put broth into flasks (straight neck and swan neck)
2. Boil to kill any preexisting microbes
3. Observe flasks for any activity

Results

Time	Straight Flask	S-Shaped Flask
0	Clear	Clear
1	Clear	Clear
2	Cloudy	Clear
3	Cloudy	Clear
4	Mould	Clear

Nutrient broth that was not exposed to microbes in air remained clear, while the broth in the straight neck flask turned cloudy, indicating spoilage. This demonstrates that microbial growth cannot arise spontaneously and is instead a result of particles in the air carrying microorganisms. This is because the curved neck flask stopped falling particles of micro bacteria contaminating the broth.

= Microbes are cause of spoilage

Conclusion - Microbes come from preexisting microbes since no growth occurred in the sterile swan-necked flask.

- assess the causes and effects of diseases on agricultural production, including but not limited to:

Role of Agriculture: form of primary industry that involves the cultivation of crops and pastures and the rearing of animals to provide meat, milk, fibres and other products for humans.

Due to Australia's isolation from the rest of the world, we are free from many infectious diseases that affect animal and plant crops in other countries. This makes our agricultural crops highly sought after in the rest of the world, increasing economic prosperity.

We must protect our agricultural industry due to transmission of many animal and plant diseases.

Exotic diseases: Exotic diseases are infectious diseases that normally do not occur in the region either because they have never been present there or because they were eradicated and then kept out by government control measures e.g. foot and mouth disease, influenza

Endemic diseases: Diseases present within a country or a region e.g. footrot, anthrax

Causes	Effect
- Loss of genetic biodiversity, due to monoculture, results in a lack of resilience to diseases - Intensive agriculture practices such as high density livestock facilitates the easy transmission of diseases (close contact) - Increased globalisation results in mobility of diseases through people carrying infected plants and food products across borders (lack of biosecurity) - Wide use of pesticides / antibiotics can result in resistant strains of diseases, increasing their prevalence and severity	- Death and illness in plants and animals - Decreased yield and quality of crops - Loss of trading opportunities both domestically and internationally resulting in a downturn in economic activity and financial hardship for farmers - Food security can be impacted e.g. potato famine - Rise in prices for fresh produce = fall in the health of the population

– plant diseases

Potato Blight:

- Potato crops were infected by a fungal pathogen (*Phytophthora infestans* fungus)
- Spreads through spores and infected tubers
- Destroyed $\frac{3}{4}$ of the entire potato crops in Ireland, poisoning 100,000 people and causing mass starvation (Irish Potato Famine)
- Was due to a lack of genetic variation in the potato population
- reminder of the importance of maintaining biodiversity in crops to minimise the transmission of disease

– animal diseases

Lyme Disease in Cattle

- Cattle are infected with the Lyme disease bacterium, *Borrelia*, through the transmission of bites from infected ticks (vector)
- dairy cattle may develop fever, stiffness and swollen joints, lameness, and their milk production decreases
- Outbreaks have been identified in high density livestock facilities aiding in the transmission of infected ticks through direct contact
- Can result in decrease in demand for cattle due to fall in quality of products and fall in quantity output
 - compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts

Pathogen - Name	Protozoan - Plasmodium (Malaria)	Bacteria - Bacillus
Facilitate Entry	Similarity Both have appendages such as flagella to help with locomotion which aids them to move through fluids. Difference Bacteria secrete cell wall degrading enzymes while protozoans have a tubular structure that	

	aids them into pushing their way into cells.
Facilitate Transmission	Similarity Both can go into a dormant state that aids them in survival outside a host cell which can be spread in the environment by wind and water to a new host Difference Protozoa can form resistant cysts that can survive for months while bacteria can form spores but not cysts

Pathogen - Name	Fungi - Potato blight / Athletes foot	Bacteria - Bacillus
Facilitate Entry	Similarity Both can produce cell wall degrading enzymes Difference Fungi hyphae growth helps them to adhere to host cells while bacteria have long pili that prevent them washing away in mucus.	
Facilitate Transmission	Similarity Both can induce produce spores to be spread and easily dispersed within an environment Difference Can induce host behaviour in different ways. Bacteria can induce coughing in order to be transmitted through air while fungus can induce organism behaviour such as dying near a damp area to ensure easy dispersal of spores.	

Virus and Bacteria Transmission (Influenza and Whooping Cough)

- Can induce host behaviour such as coughing or sneezing to aid in transmission

Protozoan and Virus Transmission (plasmodium and malaria)

- Use vectors such as mosquitoes to aid in transmission

Pathogen

Pathogen	Adaptations that facilitate entry	Adaptation that facilitate transmission between hosts
Viruses	- Viruses can either drill & insert their genetic material into the host cell - Can stimulate receptors of the cell that trigger endocytosis, and be brought directly into the cell's body - Infect vectors to enter into an organism - Can't survive outside of host	- Can change their antigens so they are not recognised as a threat by the immune system - Can spread through vectors such as mosquitoes, soil, pollen and needles - Uses host cell mechanisms to replicate (polypeptide synthesis) - Can impact host behaviour inducing coughing and sneezing, expelling virus particles that can be inhaled by others, causing spread - Can remain suspended in air for long periods of time

		without drying out
Bacteria	- Can survive outside hosts or on a dead host - Have long pili (hair) that span the thickness of mucus layers so it does not get washed away by saliva or mucus - Have fimbriae on its pili to attach to cell surfaces (adhesion)x - The bacteria helicobacter has the ability to secrete an enzyme which converts urea to ammonia which helps to combat the acidic environment - Forms spores to survive in the outside environment - Have external appendages that help them enter into cells or move to another host e.g. flagellum to move through fluid	- Have external appendages that help them enter into cells or move to another host e.g. flagellum to move through fluid - Can change their antigens so they are not recognised as a threat by the immune system - Slime layer to help it stick to surfaces - Resumes growing when conditions are more favourable - Dormant spores allows bacteria to survive in extreme conditions over long periods of time - Inducing coughing enabling pathogen to spread through air - Produce enzymes to break down and consume cell contents
Protozoa	- Uses insect vectors for entry into host - Has a tubular structure to push its way into a cell - Creates a lipid membrane once inside a host to protect against a cell's defence mechanisms - Form resistant cysts that can survive for months (thick tough wall) - Enters into privileged sites (locations within the host where the immune response cannot operate at full efficiency) - Multiply and overwhelm the immune system	- Uses insect vectors for transmission between hosts (malaria - plasmodium protozoa) - Can change the behaviour of hosts so they can be transferred to a new host e.g. rats lose fear of cats
Fungi	- Secretes cell wall degrading enzymes to break down host cell for nutrients - Adhesins that help it stick to host cells - Hyphae growth form helps fungus adhere to host tissues promoting invasive growth - Hyphae that grow into the skin or nail as dermatophytes (feed off keratin) can fragment into other growth types - Forms spores to survive in the outside environment	- Can switch between various growth forms e.g. yeast cell is useful for dispersion in the wind - Can change behaviour of host facilitating transmission e.g. zombie ants - fungi commands ant to go to a damp area before death in order for spores to spread effectively - Can survive without a host
Macro-parasites	- Secrete immunomodulatory proteins that reduce the host's immune response - Molecules in saliva prevent vasoconstriction to prevent host initiating inflammatory response - Have specialised structural adaptations e.g. hooks on their mouth or legs that allow them to piece/attach/dig on to host	- Can change host behaviour facilitating transmission e.g. rodents attracted to smell of cat urine - Faeces-Oral Transmission e.g. infected organisms can secrete tapeworm eggs which can contaminate food and water. An individual will be infected upon consuming the tapeworm larvae.

Responses to Pathogens - How does a plant or animal respond to infection?

- investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-source investigation, for example:

Plants do not have an immune system like animals do.

Passive defence mechanisms are those that are present before contact with the pathogen (physical barriers), while **active defence** mechanisms are activated only after pathogen recognition (chemical barriers) - (passive defence is not technically a response)

Passive Defence - Physical Barriers (1st line of defence) - NOT A RESPONSE (INNATE BARRIER)

- Cell walls contain lignin and cellulose or outer bark
- Waxy epidermal cuticles

Active Defence / Barriers

1. Gene for Gene resistance: some plants possess resistance genes which produce proteins that are able to disable specific pathogens - in response to entry of a pathogen

2. Basal Resistance: When a pathogen is detected, the gaps between cells shut to reduce the chance of a pathogen spreading throughout the tissue

3. Hypersensitive response: if basal resistance fails, cells surrounding the pathogen die, cutting off nutrient source for a pathogen - causes dying / yellowing leaves

4. Systemic (throughout) acquired resistance: produces salicylic acid (signally molecule) to alarm the rest of the plant tissues that a pathogen has invaded - communicates the initiation of other chemical barriers

- Stomata can be closed when signaled

– fungal pathogens

Disease: Myrtle Rust disease - Myrtle Rust Fungus - Eucalyptus

Causes: deformed leaves, defoliation, reduced fertility, stunted growth and plant death in bottlebrushes, tea trees and eucalyptus

How:

- Spores land on the stem of the plant causing the site of infection
- Spores send out a germ tube to find an opening into the plant
- Once an opening is found, the fungus penetrates deep into the plant tissue to access nutrients
- Fungus uses this nutrients to grow more fungi

Plant Response:

- Fungus releases small proteins molecules called effectors into the cell
- Plant cell can detects fungus receptors by releasing proteins that act as defence against the fungus by binding to the effector proteins
- This binding informs the plant that there is an infection taking place
- When the plant knows that there are infected cells, it will then sacrifice the cells surrounding by killing them to cut off nutrient supply to the fungus
- After cell death the pathogen will eventually starve from having no nutrients and die
- Gene for Gene response: produces chitin a chemical which breaks the fungus's cell wall

– viral pathogens

Disease: Banana bunch disease - Banana bunchy top virus

Causes: prevents plant producing fruit, reduced growth with emerging leaves becoming choked and bunched

How:

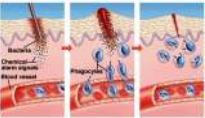
- Infected Aphids and other insects that tap into the phloem to feed act as a vector, creating a pathway for the transmission of the virus
- Leaf munching insects create a wound site for viruses to enter into a plant
- Virus invades plant cells in order to replicate more viral particles
- The virus can inhibit cellular processes such as water and nutrient uptake and even photosynthesis

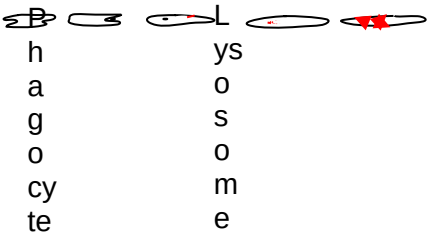
Plant Response:

- Systemic acquired response
- Hypersensitive response (evidence is yellowing leaves)

- analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animals cells and tissues

Line of Defence	Response (immune system) Occurs only when pathogen presence is detected = activated (active)	Barrier Barriers to pathogens that are always working (passive)	Innate immunity Defence that is present at birth and is non-specific	Adaptive immunity Immunity defence is specific to pathogen present
First		✓	✓	
Second	✓		✓	
Third	✓			✓

Physical barrier against infection	Physical responses to infection	Chemical barriers against infection	Chemical response to infection
Microbiome - Microbes are microscopic organisms that live on the skin, respiratory tract, intestines, mouth and vagina. - Microbes outcompete pathogens on the skin, protecting the body from disease. - Impact on homeostasis can change the microbiome, causing an imbalance in microbes which makes it easier for pathogens to survive	Vomiting and diarrhoea - Reflex to pathogens present in the stomach - Bodies way of expelling harmful substances to reduce the number of pathogens present in the body	Urine - Urinating flushes pathogens out of the body - Antimicrobial peptides are secreted by the cells lining the urinary tract to kill bacteria and prevent them from attaching to cells - pH of urine creates a good environment for phagocytes to work best in killing pathogens	Inflammation (chemical response) - When cells are challenged by pathogens or damaged, they release chemical signals (histamines) that cause the capillaries to dilate and increase in permeability (allows movement of white blood cells to infected site) - perform phagocytosis & nutrients to heal the wound) - swelling due to increased blood flow <u>Damaged cells onsets the release of chemicals (histamines) that:</u> - attract white blood cells & nutrients - Increase the bodies temperature (fever) 
Skin - Tightly packed cells that contains keratin which is hard for pathogens to break through - Epidermis is the thin outer layer of the skin which is resistant to degradation by bacterial enzymes	Increased urination When the bladder is attacked by a pathogen urine is used to flush away microbes from the urinary track (urine)	Sebum and sweat - Sebum is an oily material secreted by sebaceous glands - It waterproofs and lubricates the skin, providing a protective film against microbial growth - Sweat and other skin secretions	Phagocytosis (and chemical signalling) - The process where phagocytes / macrophages (specialized types of white blood cell) surround a pathogen and completely enclose/engulf it - A type of endocytosis - Once the foreign particle is fully inside it, enzymes destroy it.

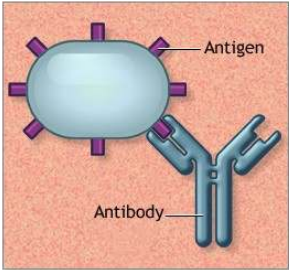
- The skin is well supplied with blood, contributing to their effectiveness in fighting infections due to easy supply of white blood cells - Constant shedding carries microbes away from the surface	is sterile)	release toxins, salt and lysozyme (enzymes) that wash, break down bacterial cell walls and kill microbes	- Lysosomes are an organelle found in white blood cells that produce digestive enzymes Antigen is displayed on the surface by phagocytes to alert 3rd line of defence 
Mucous membranes /mucous - Internal cavities are lined with epithelial tissue called mucous membrane - Mucous membrane forms a barrier to pathogen entry by trapping pathogens - Works in conjunction with cilia (tiny hairs) which beat in coordination, pushing pathogens out of the body	Wound healing - When there is a breach in the body's tissue, it becomes exposed to environmental pathogens - Platelet in the blood seal the wound to prevent pathogens entering - Protein called Fibrin forms a mesh to trap platelet to form a clot to seal the wound	Commensal Bacteria Good bacteria that secretes antibacterial compounds, killing pathogens that enter into the host. (Antibiotics are harmful to commensal bacteria)	The complement system - Group of 20 proteins that assist other defense mechanisms (second line) in destroying extracellular pathogens - They stimulate phagocytes and attract them to the site of infection where they can destroy the membranes of the invading cell - Bound antibodies activate the cascade of complement proteins which then can then breakdown the membrane of a pathogen/invaded cell causing it to burst or can attract phagocytes to destroy pathogens
Tight junctions - Blood vessels are lined with endothelial cells (semi-permeable membrane) - These cells can become compact, preventing the entry of pathogens from infected tissue into the bloodstream - If pathogens enter the bloodstream, they can travel to distant sites, causing a new infection - E.g. blood brain barrier restricts diffusion of pathogens into brain		Tears - Tear film covers the cornea & conjunctiva and contains chemical substances which have antimicrobial properties, including AMPs, lysozyme, & IgA - Lysozyme breaks down cell walls and thus acts as an antibiotic; killing bacteria - Crying work to wash away pathogens (response)	Fever (pyrogens as chemical response) - The hypothalamus regulates the body's temperature - The body may react to a pathogen by altering the hypothalamic set body temp, allowing the body to heat up - It does this by releasing 'fever causing' chemicals known as pyrogens - The heat kills and limits the growth of pathogens + enhance the activity of white blood cells + onset sweat to flush out pathogens
Peristalsis - Smooth muscle in the alimentary canal (tube where food moves from the mouth to the anus) constantly contracts in one direction to prevent intestinal bacterial growth	Granuloma / cyst - Cells die creating a wall of dead cells to seal off an infected site creating a capsule known as a granuloma - prevents infection from spreading - Cells inside the granuloma then die, destroying the pathogens that caused the infection	Gastric (stomach) secretions - Cells lining the stomach secrete hydrochloric acid - This highly acidic environment discourages the growth and survival of microbes	Cytokines - Protein chemical signals released by cells in response to pathogens to signal to neighbouring uninfected cells E.G - Cytokines (interferons) signal neighbouring cells to reduce protein synthesis to stop replication of viruses. - Cytokines onset lethargy, muscle pain and nausea to signal the animal to isolate and rest - Cytokines signal neighbouring cells to go under apoptosis (programmed death of damaged cells - e.g. cancerous cells)
Sphincters - A circular muscle that maintains constriction of natural body passages / openings - By narrowing / closing openings, the muscle physically seals off parts of the body to reduce risk of pathogen invasion		Saliva - Saliva consists of water, mucus, electrolytes, enzymes and antimicrobial substances such as lysozyme and immunoglobulin A (IgA) - Saliva can flush microbes away and kill microbes using antimicrobial peptides	Lymphatic System -Set of vessels that flow closely to the circulatory system but filter interstitial fluid - delivers lymph that carries white blood cells -Removes fluids in body tissue that leak out of blood vessels and cleans this fluid in lymph nodes (creates lymphocytes (type of white blood cell)) -Lymph system then joins back to heart delivering clean fluids -Model: Sewage system of the body

Immunity - How does the human immune system respond to exposure to a pathogen?

Antigens - a substance that induces the formation of antibodies, therefore stimulating an immune system response. Antigens are proteins on the surface of pathogens and are how the immune system recognises threats. Often invading pathogens are referred to as antigens. *The immune system doesn't affect self cells because our cells contain self antigen markers, indicating that they're not foreign particles.*

Antibodies

- Proteins which are produced in response to the presence of an antigen in the body.
- Y shaped with two binding sites



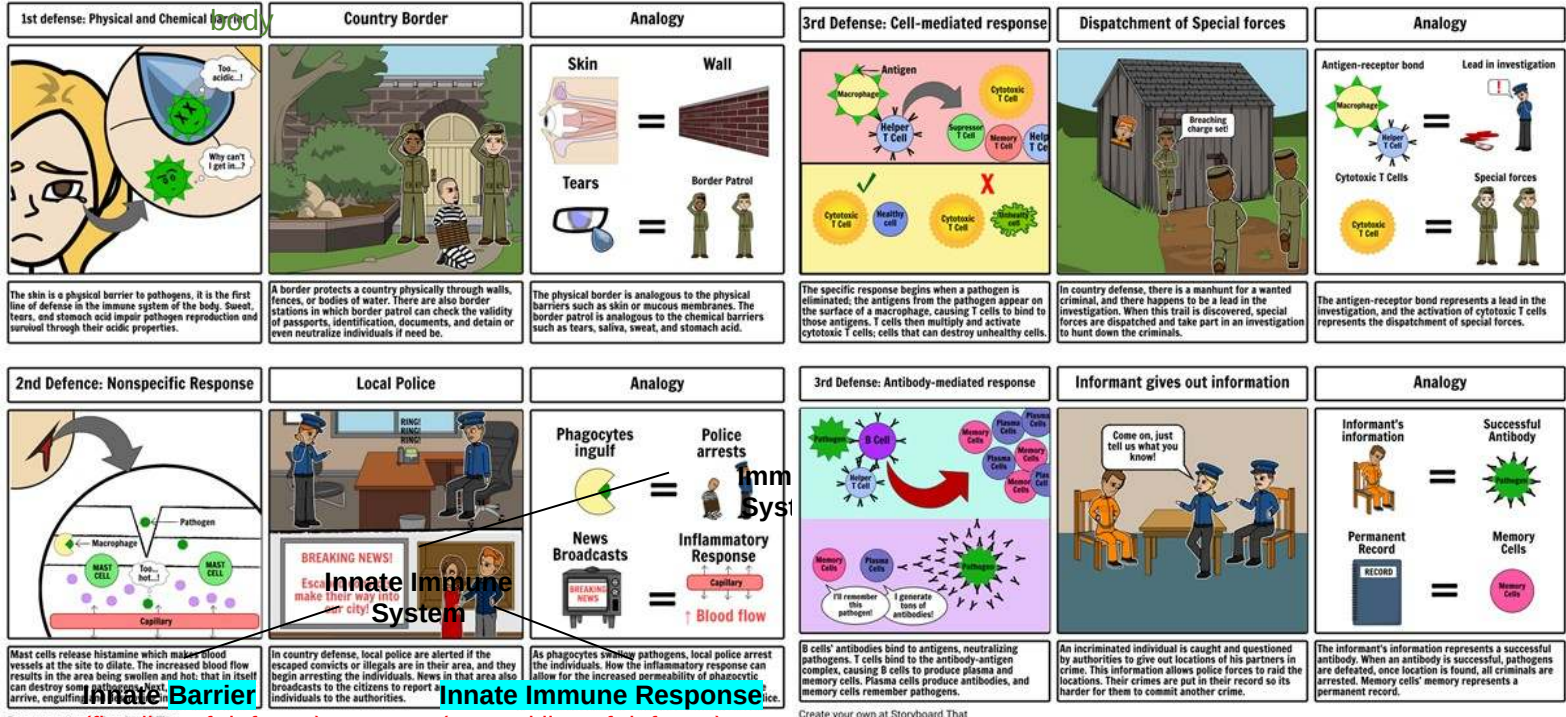
- Antibodies develop structures that recognise and bind to specific antigens
- Can be found on the surface of Plasma B-cells or in bodily fluid
- Produced from plasma B-cells when they are activated (come into contact with antigen)
- The formation of the antigen-antibody complex causes the deactivation of the antigen.

Lymph - Fluid that leaks out of capillaries into spaces between cells; circulates in the vessels of the lymphatic system.

lymph node - Small structures located on lymphatic vessels where pathogens are filtered from lymph and destroyed by lymphocytes.

lymphatic system - System of the body consisting of organs, vessels, nodes, and lymph that produces lymphocytes (in the bone marrow) and filters pathogens from body fluids.

- investigate and model the innate and adaptive immune systems in the human



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Innate Barrier
(first line of defence)
In place at birth and prevents entry of pathogens. Response is also non-specific
E.g. skin, mucous, cilia, tears, sweat, urine

Innate Immune Response
(second line of defence)
In place at birth and occurs when pathogen is detected after entering the tissue. Response is also non-specific.
E.g. inflammatory response, phagocytosis, cell death, complement system, fever, cytokines

meaning the immune system will remember subsequent encounters with the same antigen, allowing them to effectively refight the pathogen.

Cell mediated immunity
Immunity
Humoral

Innate Immune System

- Inflammation
- Phagocytes
- Lymph System

- Cell death to seal off pathogen
- The complement system
- Natural Killer Cells that kill infected cells

Adaptive/Acquired Immune System

The difference between lymphocytes and phagocytes is that lymphocytes generate specific immune responses against pathogens whereas phagocytes generate the same response to any pathogen. Lymphocytes are a type of white blood cell produced

When B and T cells bind to an antigen, they become "activated" differentiating into specific cell functions.

Cell mediated immunity

Involves T-lymphocytes (T cells)

Helper T-Cells

Gets alerted by phagocytes (engulfed a pathogen and has antigen displayed on surface) or antigens present which then causes the duplication of T-cells which release chemicals such as cytokines that alert the cloning & activation of T and B cells to differentiate into cytotoxic T-cells and B cells -> activates cell mediated immunity & humoral immunity

Memory T-Cells

Remain in the body in order to remember the antigen and respond faster on subsequent infections by the same antigen. It is these memory cells that allow the adaptive immune response to improve over time, becoming faster and stronger

Cytotoxic T-Cells

Specialise when coming in contact with an antigen and produce chemicals that destroy any cell infected by pathogen with the antigen marker

Suppressor T-Cells

Stops the immune system response after the antigen has been eliminated

Humoral / Antibody mediated Immunity

Involves B-lymphocytes (B-cells)

Plasma B-cells

Specialises when it comes into contact with antigens. It will then produce antibodies that can respond to specific antigens present.

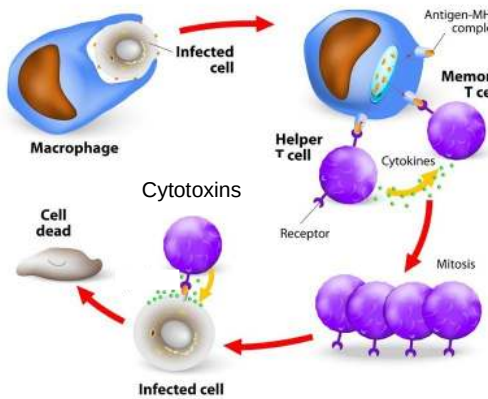
Antibodies bind to antigen and form antigen-antibody complex to inactivate antigen.

Memory B-cells

Same as memory T-cells except if it comes across an antigen it remembers, it turns into plasma cells and responds fast producing antibodies used in subsequent infections

(need both memory B and T cells as T cells can detect infected cells and B cells can rapidly produce antibodies but can't enter cells)

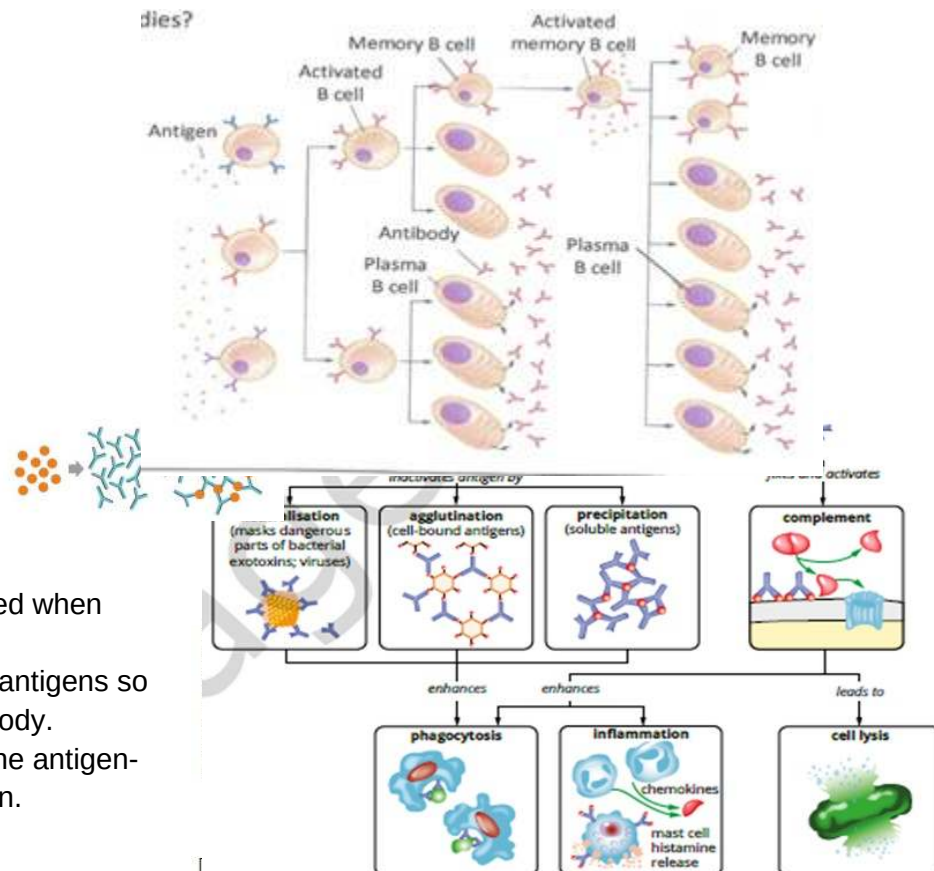
CELL-MEDIATED IMMUNE RESPONSE



How do Antibodies fight antigens?

An antigen-antibody complex is formed when antibodies attach to antigens. These antibodies disable / inactivate antigens so they are no longer able to harm the body. Phagocytes come along and engulf the antigen-antibody complex, killing the pathogen.

dies?



Adaptive Immunity

Active

Natural: when you come across

Passive

Baby gets antibodies through placenta or breast

exposure, there is a negligible amount of antibody in the system and no memory cells for this microbe.

the antibody production decreases, as fewer antigens are present. Memory cells are still present.

activate the memory B cells which quickly differentiate into antibody-secreting plasma cells producing large amounts of antibody very quickly.

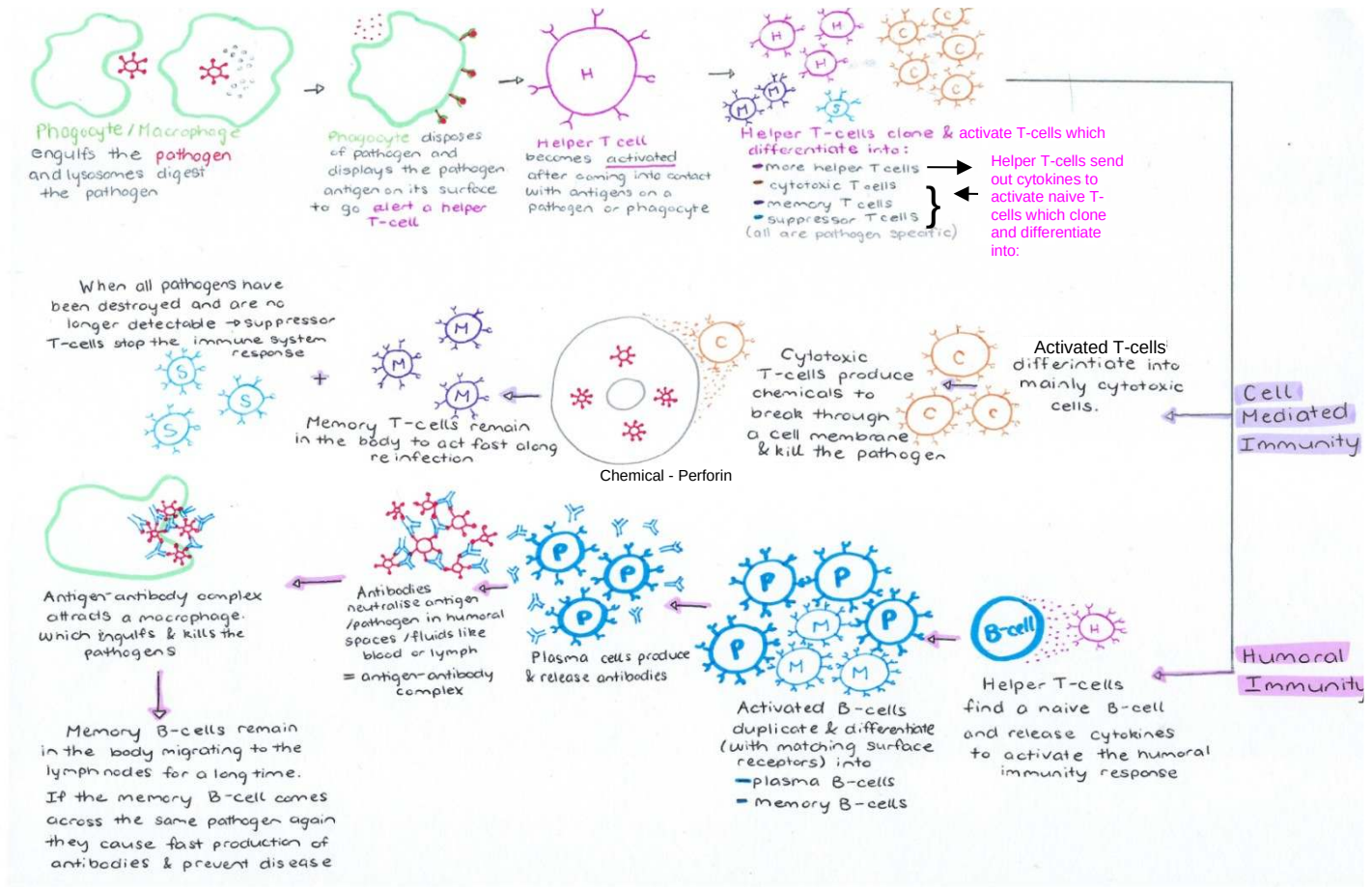
the antibody production decreases, as fewer antigens are present. A higher level of immunity is gained as more memory B and T cells are formed.

Once the body is exposed to antigens (through disease or vaccine), the B cells are activated and produce antibodies. They divide to produce plasma cells and memory cells. As the number of plasma cells increases, the concentration of antibodies increases.

body develops
memory cells
Artificial: if you
are vaccinated
and you develop
memory cells

People with
suppressed
immune systems
(organ transplant
recipients)
can get transfusion
of antibodies

Third line of defence:



- explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity (secondary exposure)

The **innate immune system responds the same way to a secondary exposure** as they have no memory and therefore does not learn anything.

Adaptive immunity Secondary Exposure:

1. Memory B (or Memory T) cell detect the presence of a memorised antigen
2. Memory B cells alert helper T-cells about re-infection
3. Memory B cells differentiate into plasma B cells which produce large quantities of antibodies fast so that signs of disease never develop
3. Helper T cells are activated, differentiating into cytotoxins to kill off pathogens (cell mediated response)

Prevention, Treatment and Control - How can the spread of infectious diseases be controlled?

There are many laws and legislations that have been put in place to control the spread of infectious disease.

- Laws can improve access to vaccinations and contraceptives, and by facilitating screening, counselling and education of those at risk of infection. Law also has a reactive role: supporting access to treatment, and authorizing public health authorities to limit contact with infectious individuals and to exercise emergency powers in response to disease outbreaks
- Immunisation is a successful and cost-effective public health strategy that saves millions of lives each year. Governments can support vaccination coverage by ensuring that vaccination is free or affordable
- Screening individuals to determine if they have been infected with or exposed to an infectious disease is a core public health strategy. E.g. screening individuals for COVID
- Laws can ensure the confidentiality of test results, and protecting individuals diagnosed with particular diseases from discrimination.
- Australia has strong biosecurity laws surrounding the entry and exit of goods
- Public health laws should authorize compulsory treatment only in circumstances where an individual is unable or unwilling to consent to treatment, and where their behaviour creates a significant risk of transmission of a serious disease
- Public health laws may authorize the isolation of individuals and groups who may have been exposed to an infectious disease, as well as the closure of businesses and premises and the confiscation of property e.g. during COVID

- investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease

Malaria: vector (mosquito) infectious disease causing fever symptoms.

Geographical Section	Measures put in place	Judgement of effectiveness
Local - country (preventative)	- Wearing insect repellent - Sleeping with insecticide treated mosquito nets - Indoor residual spraying - Saying in places with effective air conditioning and screening on doors and windows - Educating people about the disease - Taking anti-malaria medication - Killing mosquitoes on a large scale using pesticides - Increasing the number of healthcare facilities to treat people with malaria - Drainage programs to get rid of still water where mosquitoes can breed	- Protective measures prevent the increase in reservoir size, decreasing the ability for mosquitoes to pass the disease - Educating individuals increases awareness and participation in protective behaviours. It also increases rates of early detection and treatment. - Killing mosquitoes is effective as they are the source of transmission
Regional - groups of countries (control)	- Wearing long sleeves and covered clothing when travelling - Genetically engineering mosquitoes which mate with malaria vectors, passing a fungus that kills them (CRISPR-Cas9 gene-editing technology was used to make changes in mosquito genes that could reduce their ability to spread malaria. This could provide a new way to reduce illnesses and deaths caused by malaria.) - Establishing government monitoring programs to better react to an outbreak - Testing travellers across borders - Vaccinations - Insecticides - Maintained drains to prevent breeding sites - Spray illegal boats entering the country - Spraying planes with insecticides	- Genetically engineering mosquitoes can be highly effective in eradicating vectors before transmission as increasing numbers of mosquitos are becoming resistant to pesticides and malaria pathogens are getting resistant to antimalarial drugs. - Creating a government strategy can reduce the blowout of the outbreak and contain it faster before it spreads - Australia and many other countries used an insecticide called DDT which killed mosquitoes, stopping transmission = effective - Last 3 methods have been effective in Australia as they have eradicated mosquitoes carrying this pathogen
Global (control)	- Media awareness campaigns - Closing flights in and out of countries - Closing off travellers from malaria endemic areas - Informing and vaccinating travellers	- Testing travellers and closing borders will prevent the spread of the disease into new population groups

- Screening people entering the country

- **investigate procedures that can be employed to prevent the spread of disease, including but not limited to:** (Link to a reduced number and chance of pathogens to be spread & enter body)

– hygiene practices

Some microbes use human waste products as sources of nutrients. Microbes can live on food that has been left in a warm place for some time, in water supplies that have been contaminated with waste (for example sewage) and in human faeces. Because food, water and personal hygiene habits are potential sources of infection by pathogens, it is important that they are kept sterile or disposed of in a hygienic manner. Heat, detergents and water, used to achieve cleanliness, kill or remove pathogens.

Hygiene

Food	Water	Personal
- Wash fresh produce thoroughly - Prevent cross contamination (raw meat) - Keep hot foods above 60 degrees - Cover and seal food for storage - Fully cook pink meat to kill all pathogens - Strict hygiene guidelines at restaurants	- Water purification system - Boiling water before use in third world countries - Domestic water supply in Australia must comply with strict standards	- Disinfecting cuts and covering them up - Regularly brushing teeth - Washing hands frequently (before eating, after blowing your nose etc.) - Washing dishes with detergent

These practices all aim to kill pathogens or reduce their presence to decrease the chances of pathogens entering the body.

– quarantine

Quarantine: isolation of a person who has been in possible contact/exposed to an infectious disease. Not necessarily have the disease.

It is a biosecurity measure that allows us to keep diseases we don't have in the country out of the country. This is because the movement of people, animals and plants, across borders, can carry pests, disease and microbes across borders. Quarantine periods differ depending on the pathogen as they have different incubation periods. E.g. 19 days for COVID

It is an effective measure that prevents further transmission of the disease.

- Protects all borders: water, exports/imports, immigration ect.

Example: dogs at borders sniff for fresh fruit. This is before fresh produce can carry Mediterranean fruit flies which are detrimental to the Queensland banana industry.

– vaccination, including passive and active immunity

Immunisation: a process which stimulates the immune system to produce memory B and T cells and antibodies so that on a secondary infection, the body can effectively fight off an infection.

Vaccination is a method of providing immunisation, in which weakened/dead or replicated pathogens are introduced to the body.

Vaccination: a chemical preparation that contains antigens which induce the third line of defence. This results in the body making antibodies and memory B and T cells. When the body comes across the real pathogen, the body has immunological memory and can effectively fight off the disease through a quick response.

Active immunity - immunity developed by undergoing the immune response (memory B and T cells)

Passive immunity- immunity gained by obtaining antibodies from a vaccination or through the placenta.

Only lasts a few months as the body is actively making antibodies.

(vaccines can be passive and active)

Notes: antibodies given to a newborn baby do not last forever, therefore passive immunity wears off

Herd Immunity - form of immunity that occurs when the vaccination of a significant portion of a population provides a measure of protection for individuals who have not developed immunity

– public health campaigns

Programs that are designed to reduce the spread of diseases.

Examples:

- Government regulations and standards for handling, cooking and storing food in kitchens, water quality and sterilising equipment at clinics and hospitals
- Awareness campaigns have been used during COVID to raise awareness on the signs and symptoms and importance of getting tested when sick
- Laws require certain diseases to be reported to authorities if they are detected. This allows early detection in order to control the spread of the disease.
- Legislations during COVID required individuals to stay inside and only go out for essentials which required wearing a mask. This limited human contact to stop the spread of the disease.
- Public immunisation programs (commonly done as a child through school nurse visits) for diphtheria, tetanus, whooping cough, measles, mumps and rubella help to prevent contracting these diseases. This also increases herd immunity, reducing the transmission of the disease.
- Public Education programs are put in place to educate individuals on protective factors towards their health. E.g. as part of the highschool curriculum children are educated on safe sex to reduce the spread of chlamydia, HIV, AIDs.

– use of pesticides

Pesticides can be used to kill the vectors that can carry harmful pathogens. Killing these vectors reduces the occurrence of the disease through preventing its mode of spreading. Pesticides include insecticides (insects), fungicides (fungus) and herbicides (weeds).

E.g. mosquitoes are vectors for malaria and dengue fever. The chemical DDT was used to kill these mosquitoes to prevent the spread of the disease.

Pesticides are also used widely in Australia to spray imported items into Australia to kill any insects or other organisms present.

- genetic engineering

Genetic engineering involves altering the genetic composition of an organism.

- Altering the genetic makeup of an organism can result in the production of disease resistant plants and animals to prevent the occurrence of disease and therefore control its spread within a population e.g. **BT cotton resistant to caterpillars**
- Genetic engineering can lead to the creation of vaccines that immunise large population groups, preventing the spread of a disease
- Can also be used to create pesticides to control pathogen numbers and spread
- Can be used to alter vectors e.g. **mosquitoes** have been altered so they are unable to transmit the disease

Mortality from infectious diseases has declined significantly in many parts of the world over the past century. This has been due to four major factors

- **increased hygiene (to prevent disease),**
- **widespread immunisation procedures (to control/prevent disease),**
- **improved nutrition (to resist disease)**
- **the use of antibiotics & antiviral medication (to treat disease).**

- **investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example:**

– antivirals

Antiviral drugs are prescription medicines that fight against viruses in the body. They inhibit the development of viruses inside a cell, allowing the body's natural defense system to kill the virus. Viruses most commonly targeted include: HIV, influenza, herpes and hepatitis B and C.

They do not destroy viruses, rather inhibit their replication.

They target only 1 particular virus and can be used as a preventative medicine.

Example: There are 3 types of antivirals used for HIV/AIDS:

Drug	PrEP - Pre-exposure prophylaxis for people not infected but live with someone who is infected. Prevents a virus from establishing itself in the body.	PEP - Post Exposure Prophylaxis is only effective in the first 48-72hrs. It is a medication designated to prevent HIV infection if you have been exposed to the virus.	ART - Antiretroviral therapy, for someone who is infected. They would take this daily medicine to reduce the daily count of a virus to prevent the transmission of this disease. It is very effective in decreasing the viral load down to an undetectable level.
Effectiveness	Decreases in HIV Incidence Coincide with Expanded PrEP Use Retrospective study from the research database of Clinique médicale l'Actuel, 2011-2016 Figure 1: Trends in HIV testing, diagnoses and prophylactic programs Drugs have been effective as the the number of HIV diagnoses rates have gone down. The mortality rate from HIV has reduced for patients taking ART therefore is effective. Numbers of Patient vs Years		
Cons	Cons: these medications are very expensive and often are not subsidized by the government. Therefore it is out of reach for developing countries and lower socioeconomic groups where HIV and AIDS is of greater impact due to lack of education. Additionally developing safe and effective antiviral drugs is challenging as viruses use the host cell's machinery to reproduce. Therefore drugs used to target a virus often cause harm to the host cell. Another challenge is that there is extreme diversity between viruses, some have DNA, some have RNA, some single stranded, some are double. This makes it hard to produce broad spectrum antivirals.		

– antibiotics

Antibiotic: chemical capable of destroying or inhibiting the growth of bacterial cells. There are a range of antibiotics to treat a whole range of bacterial diseases. Antibiotics are produced naturally by fungi or soil bacteria or artificially synthesised in the lab. They work by:

- Interfering with the metabolic activity of bacteria
- Damaging the cell membrane of bacteria

Example: Penicillin

Penicillin attacks enzymes that build the cell walls of bacteria. Therefore the drug greatly weakens the cell wall, increasing the permeability for water to flow in. This causes the bacteria to swell and explode, causing the bacteria to die and allowing a person to recover from a bacterial infection.

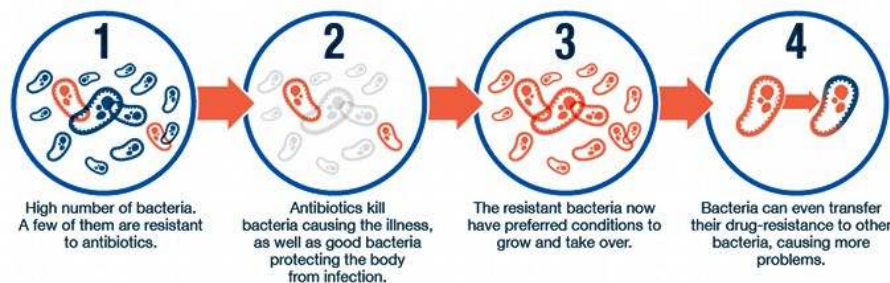
Effectiveness - Estimates are that 200 million lives have been directly saved from the discovery of penicillin.

Genetic technology:

Cons

- Bacteria can grow resistant to antibiotics, reducing the effectiveness of antibiotics in controlling the outbreak of a disease. Antibiotic resistance leads to higher medical costs, prolonged hospital stays, and increased mortality. This can be due to the overuse / unnecessary use of antibiotics, making resistant bacteria more common.

How does antibiotic resistance occur?



- Individuals may fail to complete the full prescription course, resulting in bacteria surviving and redeveloping an infection
- People can be allergic to antibiotics e.g. penicillin
- Antibiotics only work on bacterial infections
- Reducing habits and reservoirs of vectors
- Draining water reservoirs where mosquitoes can breed
- Blocking up drains and pipes where bats may live

- investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic

Ebola Virus Case Study

Methods of control have been effective as ebola was contained to a small region in africa and reduced to a point where it is no longer a pandemic.

Evaluate - link management strategy to reduce chance of contracting and spreading the disease, reducing incident rates

Environmental Management		
Control	Evaluate	Cons
- Use of protective gear by health staff such as full body suits, gloves, face shield to prevent contact. - Frequent cleaning of infected surfaces - Training doctors and nurses on health and safety protocols such as sanitising equipment - Safe and prompt burials using protective clothing to prevent coming into contact with the infected body. Trusted community members should deliver these messages. - Educating individuals about eating wild meat as they can be easily contaminated and pass on the virus - Educating individuals on protective health behaviours such as regular hand washing, fully cooking meat, cleaning all surfaces, sanitising water (boiling, importing), properly seal food, disposal of infected items and properly disposing of rubbish - Proper provision of adequate and hygienic facilities for barrier nursing and treating patients - Limiting deforestation as it increases density of animals, increasing the spread of the disease - Isolation and treatment centres - Contact tracing involves identifying, testing and quarantining all individuals who have potentially come into contact with infected individuals. - constant monitoring	Personal protection gear prevents contaminated body fluids coming into contact with healthy people, reducing the spread of the disease and decreasing incidence numbers. Educating individuals on health protective behaviours reduces the number of individuals exposing themselves to the virus, reducing the spread of the disease and decreasing incidence numbers.	- Traditions are important to communities and therefore it will be hard to change behaviors. Burial restrictions limit people's cultural freedoms - Some groups rely on wild meat as a source of food and therefore will not change their behaviours - Often it is hard to maintain health protecting behaviours in a developing country where there is a lack of funds for sanitary and water infrastructure - This conflicts economic growth and development such as building of infrastructure to increase the quality of life and creating new jobs - Difficult to monitor human behaviour
Quarantine		
Control	Evaluate	Cons
- Quarantining individuals until they either show signs of the disease or are no longer at risk (42 days - double incubation period) - Placing sick people in beds 3m apart to reduce contact - Assigning clinical staff to particular patients to reduce potential for transmission - Isolating infected people in hospitals and ebola virus centres - Quarantining individuals moving in and out of countries - Screening all individuals moving across borders	This prevents the contact between infected people and non-infected people, since ebola is transmitted through body fluids. Reduced contact leads to reduced change of spread which decreases the prevalence and incident rates of the disease.	- Unethical to quarantine people as it limits their freedom - Difficult to monitor mobility of all people

- interpret data relating to the incidence and prevalence of infectious disease in populations, for example:

Incidence: the number of new cases

Prevalence: total number of people who have the disease (all active cases + new cases)

Epidemic: A sudden increase in the prevalence of a disease above the endemic rate in a country

Pandemic: A sudden increase in the prevalence of a disease across multiple countries and/or multiple continents.

Outbreak: A sudden increase in the prevalence of a disease above the endemic rate within a small Or specific geographic area.

Endemic level of disease: The baseline rate at Which a specific disease typically occurs in a given population e.g. chicken pox in Australia

– mobility of individuals and the portion that are immune or immunised

Humans act as a carrier for pathogens therefore population mobility increases the chance of the spread of disease. Although some diseases that are infectious, we have immunity against them (vaccinations). For some diseases there are no vaccines.

Herd Immunity: when an estimated 80% of a population is vaccinated against a disease, there is less of the disease in the community, making it harder for the disease to spread. This is essential to protect individuals who cannot be vaccinated for medical reasons or babies who are too young to yet receive vaccinations. Therefore the decision to not vaccinate can therefore affect other individuals, even kill them.

Even though diseases may be eradicated from Australia, we must still immunise for these diseases due to high mobility where a disease can be carried back into the country. Good herd immunity means that the community transmission is reduced to prevent the spread of the disease.

An effective vaccine reduces the incidence of the disease. It is a preventable measure not a cure as it requires activating your immune system response. Therefore to determine the effectiveness of a vaccine, it is important to look at the incidence of disease, not the mortality.

– Malaria or Dengue Fever in South East Asia

(both mosquito borne infections that cause flu-like symptoms and can be fatal)

Why are these diseases prevalent in tropical and subtropical regions?

- Tropical areas - high rainfall and temperature is ideal breeding conditions for mosquitoes
- Less infrastructure and funds to reduce reservoir



Dengue Fever - on the rise

- Due to increased urbanisation in these regions, half of the world's population is now at risk of contracting dengue fever
- Prevalence of the disease is on the rise -> we have learnt how to diagnose dengue fever and a lot of people don't see the doctor as they mistake the disease for the flu. This allows the disease to spread as mosquitoes spread the disease from unknown infected people to uninfected people. Spread is therefore due to misdiagnosis

Malaria - gradually falling

- Decrease in incidence numbers in south east asia which has lead to a drop in mortality rates
- Currently eradicated all malarious mosquitoes in Australia (still cases though)

- evaluate historical, culturally diverse and current strategies to predict and control the spread of disease

Historical	Louis Pasteur - Predicted that diseases spread through contaminating sterile objects - He invented the process of pasteurization to kill microbes present in goods in order to reduce pathogen exposure and entry Robert Koch - Urged for the sterilization of surgical instruments using heat to reduce spread of disease
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	- Created Koch's postulates to identify the cause of infectious diseases - Developed pure culture (pure isolation of microbes) which helped to understand how a microbe caused a disease John Snow - Proposed that cholera was water borne through identifying the trend that higher prevalence of cases occurred around specific water pumps - Removed water pump that was responsible for harbouring contaminated water - Effective as educated people on the importance of water sanitation Quarantine - Quarantine practices can be dated back to the ancient hebrews who would separate lepers (people with leprosy) from society as they were considered unclean - Quarantine practices were used during the black plague where ships had to wait 40 days before docking into venice Bloodletting - 19th century doctors believed draining blood could treat any illness or disease. - Was ineffective as blood is essential for transporting nutrients (most importantly white blood cells) around the body and removing waste - Bloodletting therefore hindered the immune system response through reducing white blood cells - Also exposed other people to blood that may have been contaminated with the pathogen
Cultural	Traditional Aboriginal Medicines - Aboriginals have acquired extensive knowledge on using flora and fauna to treat diseases - They found that certain plants contain antibacterial properties and painkiller properties - Were effective in treating illnesses but not in controlling the spread Tiger parts in Chinese traditional medicine - No evidence to support the hypothesis of consuming tiger parts to assist in controlling disease spread - Threatens the extinction of tiger species - People who believe in this medicine will ignore seeking appropriate help - Therefore highly ineffective in controlling the spread of disease
Current	Global Surveillance of health cases (similar to John Snow) - Healthcare providers are required to report all cases of specific diseases to the Australian Department of health (has been made easier through technology) - This allows for early detection, identification of the source and data analysis of health trends to create effective responses to outbreaks - E.g. during COVID tracking movement of people through sign in helped to identify close contacts to prevent further spread Health Campaigns - Educate people on the importance of hygiene - E.g. during COVID wearing masks, washing hands and getting tested if signs and symptoms appear Quarantine & Isolation - Individuals who have the disease are isolated from society until they no longer harbor the pathogen - This prevents the transmission of the pathogen, controlling the spread Antibiotics & Anti-viral drugs. - Help to prevent the growth and kill bacteria + viruses therefore preventing transmission to control spread Vaccines - Strengthens the immune system through activating memory B and T cells - Prevents pathogens infecting new hosts, reducing transmission and controlling spread of pathogen - Herd immunity can protect individuals not vaccinated Viruses being used as vectors - Genetically engineering viruses to inject beneficial DNA into organisms can provide organisms DNA to fight of infections that require this gene

- investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia
- bush medicine: Traditional methods of treating and managing illnesses by indigenous Australians

Soap Tree

Crushed up soap tree leaves and water produce a soap solution that has antibacterial properties and thus works as an antiseptic. The leaves contain an acid which suppresses bacterial growth. Traditionally this was used to clean cuts on their skin.

Modern Application

The bark of soap trees are crushed and boiled into an ash that is used in a paste to relieve muscle or body pains.

FreshWater Mangrove (*barringtonia acutangula*)

Traditionally bark placed on wounds to relieve pain.

Modern Application

Traditional knowledge provided the initial observations and the inspiration for research into analgesics extracted from *barringtonia acutangula*. Griffith University investigated *B acutangula* to identify and isolate the active analgesic ingredient to be used in an analgesic drug, used as a painkiller.

Tea tree oil and Eucalyptus Oil used as a disinfectant

Aboriginal people would inhale the oils of tea tree leaves to treat coughs, crushed and applied the leaves as a paste to wounds and even brewed the leaves in tea for sore throats.

Modern Application

Tea Tree oil is used as an ingredient in many health products. It is used to treat acne and fungal infections due to its antiseptic property and anti-inflammatory properties.

– smoke bush in Western Australia

Smokebush is a plant used by Aboriginal people for their healing properties in increasing the rate of healing for cuts on their skin and other skin conditions.

Experimental results obtained from trials performed by the American National Cancer Institute revealed that the smokebush had properties capable of killing the HIV virus that can be used to cure AIDS and curing cancer.

Aboriginal people did not receive recognition for the healing capacity of the plants when the Australian government granted a US firm rights for exclusive use of the plant. The Aboriginal people also did not receive money when products derived from the plant were sold. Moreover, Aboriginal people are restricted from using their cultural knowledge due to the patent granted to the biotechnology company.

Therefore the US firm stole an endemic plant from Australia and shared the knowledge with them. No royalties or other compensation or even acknowledgement were given to the aboriginal people of WA Highlighting the shortfall in patent laws and need to protect traditional knowledge about remedy used by aboriginal people. This is why now all aboriginal IP rights are protected by law.

Why is recognition and protection of Indigenous cultural and intellectual property important?

Aboriginal knowledge of bush medicine is important in terms of culture and intellectual property (IP). ATSI people are the original founders of Aboriginal bush medicine and therefore deserve the recognition and acknowledgement of their ownership and findings. For aboriginal people, bush medicine is not only linked to physical wellbeing but also to the social, emotional and cultural wellbeing of the community = must be respected. Some bush medicine has the potential for large-scale modern pharmaceutical application, but Aboriginal people's IP must be recognised and protected so their expertise is neither disregarded nor exploited for commercial and financial use. Therefore traditional knowledge must be respected and must be acknowledged. Although we must first research and confirm the traditional remedies before medical use and recognise the IP rights of aboriginal people.

- **Bush food**

Kakadu plum is a small purple fruit which contains substantially more Vitamin C than an orange. It has been used by Aboriginal and Torres Strait Islander people to boost the body's nutrient levels. It has also been used to combat or prevent diseases such as scurvy. Similarly, the Kakadu plum is incorporated into modern natural medicines as a Vitamin C source. It is also used in skin treatments for its antioxidant properties which help restore the skin membrane and elasticity.

Grass Trees (Xanthorrhoea) are a type of tree that looks like a bush growing on top. The nectar produced by flowers can be harvested and eaten. Seeds can be crushed to make flour. The soft white leaf base also contains 5% sugar which can be eaten and has a sweet nutty taste. The resin produced can be used as glue to make weapons and the flower spikes can be used on spears.

Validity = reliable, relevance to research/tested hypothesis, controlling variables, adding a control group

Reliability = larger sample size, repeating experiment, concurrence with other research, long period of time

Spots = fungus

Bacteria = rotting

Virus = mosaic pattern

Bar graph - when data is not connected

Line graph - when data from each year is connected